A NEGATIVE ANSWER TO A QUESTION ON MONOTONE MAXIMAL CHAINS IN FINITE GROUPS

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Abstract. In an email dated August 8, 2026, V. S. Monakhov communicated a question proposed jointly with I. L. Sokhor: whether every finite group has an unrefinable subgroup chain whose successive indices are nondecreasing. We give a negative answer, even among soluble groups. For every odd prime p, we construct a soluble matrix group $G_p \leq \mathrm{GL}_5(p)$ of order 2^6p^8 with no such chain. Thus the counterexamples form an infinite family; its smallest member, G_3, has order 419904. The construction is modeled on an example of Kohler, and the obstruction follows from three elementary maximal-subgroup calculations.

1. Introduction

Let $G \neq 1$ be finite. An unrefinable subgroup chain

$$1 = G_0 < G_1 < \cdots < G_n = G$$

is one in which G_{i-1} is maximal in G_i for every i. Associate to the chain the indices $j_i = [G_i : G_{i-1}]$. We call the chain increasing if

$$j_1 \leq j_2 \leq \cdots \leq j_n.$$

In the terminology of Monakhov and Sokhor [2], this is a ($<$)-chain. In an email to the author dated August 8, 2026, V. S. Monakhov wrote that he and I. L. Sokhor proposed the following question, stated here in equivalent terminology.[1]

> Does every finite group contain an increasing unrefinable subgroup chain?

We prove the following negative answer.

Theorem 1. *For every odd prime p, there is a soluble group G_p of order 2^6p^8 which has no increasing unrefinable subgroup chain.*

The smallest group in the family has order $2^63^8 = 419904$; no claim of global minimality is made. Our construction is modeled on the $t = 2$ case of Kohler's work on maximal-subgroup indices [1]. Kohler used his construction to show that a group whose maximal subgroups have prime or prime-square index can contain subgroups with arbitrarily large maximal-subgroup index. In the matrix model below, the four coordinates in x and y play the role of four degree-one generators, while the four coordinates in z are the independent cross-commutator coordinates; consequently the associated p-group has order $p^{4+4} = p^8$. Here we use the nested index pattern itself.

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[1] This email is the source of the question. Reference [2] is cited for terminology and background, not as a published source of the universal-existence question.

2. A lemma on maximal subgroups

For a finite group A, write

$$\mathcal{I}(A) = \{[A : B] : B \text{ is maximal in } A\}.$$

Lemma 2. *Let V be an elementary abelian p-group, let H be a p'-group acting on V, put $K = V \rtimes H$, and let $\pi : K \to H$ be the natural projection. Every maximal subgroup of K is of one of the following forms:*

1. *$\pi^{-1}(J) = V \rtimes J$, where J is maximal in H;*
2. *a V-conjugate of $W \rtimes H$, where W is a maximal proper H-submodule of V.*

Proof. Let T be maximal in K. If $V \leq T$, the claim follows on passing to $K/V \cong H$. Suppose $V \nleq T$. Then $VT = K$. Put $W = V \cap T$. The group T normalizes W, while V centralizes it, so $W \lhd K$. In K/W, the groups T/W and WH/W are complements to V/W. Schur–Zassenhaus makes these complements conjugate by an element of V/W. Thus T is V-conjugate to $W \rtimes H$, and maximality of T is equivalent to maximality of W among proper H-submodules of V. □

In particular, if H is a 2-group, subgroups of the first type have index 2. If V is completely reducible, subgroups of the second type have indices equal to the orders of the irreducible quotients of V.

3. The counterexample family

Fix an odd prime p and put $F = \mathbb{F}_p$. Let

$$s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad t = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad D = \langle s, t \rangle \leq \mathrm{GL}_2(F).$$

Then $D \cong D_8$, where D_8 has order 8. The natural FD-module $U = F^2$ is absolutely irreducible. Indeed, over an algebraic closure the only t-invariant lines are the two coordinate lines, and s interchanges them.

Let $H = D \times D$. Define

$$L = \left\{ \ell(x, y, z) = \begin{pmatrix} I_2 & x & z \\ 0 & 1 & y \\ 0 & 0 & I_2 \end{pmatrix} : x \in F^{2\times 1},\ y \in F^{1\times 2},\ z \in F^{2\times 2} \right\}.$$

Here the block sizes are $2, 1, 2$. Direct multiplication gives

$$\ell(x, y, z)\ell(x', y', z') = \ell(x + x', y + y', z + z' + xy'). \tag{1}$$

With the convention $[a, b] = a^{-1}b^{-1}ab$, it follows that

$$[\ell(x, y, z), \ell(x', y', z')] = \ell(0, 0, xy' - x'y). \tag{2}$$

Thus $|L| = p^8$. Put

$$Z = \{\ell(0, 0, z)\}, \quad X = \{\ell(x, 0, 0)\}, \quad Y = \{\ell(0, y, 0)\}.$$

The group Z is central in L. The outer products xy span $F^{2\times 2}$, so (2) gives $L' = Z$. Also L has exponent p: explicitly,

$$\ell(x, y, z)^p = \ell\left(0, 0, pz + \binom{p}{2} xy\right) = 1.$$

The standard Frattini formula for a finite p-group (equivalently, the Burnside basis theorem) now gives

$$\Phi(L) = L^p L' = Z. \tag{3}$$

Embed H in $\mathrm{GL}_5(F)$ through the block-diagonal matrices $\mathrm{diag}(A, 1, B)$, with $A, B \in D$. Conjugation acts by

$$x \mapsto Ax, \qquad y \mapsto yB^{-1}, \qquad z \mapsto AzB^{-1}.$$

It therefore normalizes both L and Z, so in particular $Z \lhd G_p$ for the group

$$G_p = L \rtimes H \leq \mathrm{GL}_5(p).$$

We have $|G_p| = p^8 \cdot 8^2 = 2^6 p^8$. Both L and H are soluble, so G_p is soluble.

Set $\overline{X} = XZ/Z$ and $\overline{Y} = YZ/Z$. As an H-module,

$$L/Z = \overline{X} \oplus \overline{Y}, \tag{4}$$

where $\overline{X}$ and $\overline{Y}$ are nonisomorphic irreducible modules of dimension 2. More explicitly, the first factor of $D \times D$ acts naturally on $\overline{X}$ and the second acts trivially, while the second factor acts naturally and dually on $\overline{Y}$ and the first acts trivially. Thus

$$\ker_H(\overline{X}) = 1 \times D, \qquad \ker_H(\overline{Y}) = D \times 1,$$

which also proves that the two modules are nonisomorphic. Since $p \nmid |H| = 64$, Maschke's theorem applies. Hence the only maximal H-submodules of L/Z are $\overline{X}$ and $\overline{Y}$.

Moreover,

$$Z \cong U \boxtimes U^* \tag{5}$$

is the external tensor product for the two direct factors of H. Hence Z is an irreducible H-module of dimension 4: over an algebraic closure $\overline{F}$, both $U_{\overline{F}}$ and $U^*_{\overline{F}}$ remain irreducible, and their external tensor product is irreducible for $D \times D$.

4. The index obstruction

Lemma 3. *The following maximal-index spectra occur:*

$$\mathcal{I}(G_p) = \{2, p^2\},$$
$$\mathcal{I}(M_X) = \mathcal{I}(M_Y) = \{2, p^2, p^4\},$$
$$\mathcal{I}(N) = \{2, p^4\},$$

where

$$M_X = (Z \times X) \rtimes H, \qquad M_Y = (Z \times Y) \rtimes H, \qquad N = Z \rtimes H.$$

Every maximal subgroup of G_p of index p^2 is conjugate to M_X or M_Y. Every maximal subgroup of M_X or M_Y of index p^2 is conjugate to N.

Proof. First, every maximal subgroup M of G_p contains Z. If not, then $G_p = ZM$, and the modular law and (3) give

$$L = L \cap ZM = Z(L \cap M) = \Phi(L)(L \cap M).$$

Recall the Frattini fact that $L = \Phi(L)K$ for a subgroup $K \leq L$ implies $K = L$: otherwise a maximal subgroup of L containing K would also contain $\Phi(L)$ and hence L. Applying this with $K = L \cap M$ forces $L \cap M = L$, which contradicts $Z \not\leq M$.

It follows that the maximal subgroups of G_p are the inverse images of the maximal subgroups of

$$G_p/Z = (\overline{X} \oplus \overline{Y}) \rtimes H.$$

Apply Lemma 2 and (4). Maximal subgroups coming from H have index 2, and those coming from either irreducible summand have index p^2. This proves the first spectrum and the assertion about M_X, M_Y.

In M_X, the normal elementary abelian subgroup $Z \times X$ is a direct sum of nonisomorphic irreducible H-modules of dimensions 4 and 2. Its only maximal H-submodules are therefore its two summands. Another application of Lemma 2 gives indices $2, p^2, p^4$; the index-p^2 case removes the X

summand and gives a conjugate of N. The same argument applies to M_Y. Finally, Z is irreducible of dimension 4, so Lemma 2 applied to $N = Z \rtimes H$ gives $\mathcal{I}(N) = \{2, p^4\}$. $\square$

Proof of Theorem 1. Suppose

$$1 = G_0 < G_1 < \cdots < G_n = G_p$$

is unrefinable and its indices $j_i = [G_i : G_{i-1}]$ are nondecreasing. By Lemma 3, $j_n \in \{2, p^2\}$. We cannot have $j_n = 2$: then every j_i would be 2, whereas p divides $|G_p|$. Thus $j_n = p^2$, and G_{n-1} is conjugate to M_X or M_Y.

Now $j_{n-1} \le p^2$. The middle spectrum in Lemma 3 shows that j_{n-1} is 2 or p^2. It cannot be 2, for the same order argument applied to G_{n-1}. Hence $j_{n-1} = p^2$, and G_{n-2} is conjugate to N.

Finally, $j_{n-2} \le p^2$, but $\mathcal{I}(N) = \{2, p^4\}$. Therefore $j_{n-2} = 2$. This forces all preceding indices to equal 2, impossible because p divides $|N|$. The contradiction proves the theorem. $\square$

5. The fully explicit group G_3

For ease of independent verification, we spell out the smallest example without parameters. All entries in this paragraph lie in $\mathbb{F}_3$, so that $-1 = 2$. Write $e_{ij} = I_5 + E_{ij}$ and set

$$a_s = \operatorname{diag}(s, 1, I_2), \quad a_t = \operatorname{diag}(t, 1, I_2),$$
$$b_s = \operatorname{diag}(I_2, 1, s), \quad b_t = \operatorname{diag}(I_2, 1, t).$$

Then the concrete example is the following subgroup of $\mathrm{GL}_5(3)$:

$$G_3 = \langle e_{13}, e_{23}, e_{34}, e_{35}, a_s, a_t, b_s, b_t \rangle. \tag{6}$$

Define

$$\begin{aligned} Z_3 &= \langle e_{14}, e_{15}, e_{24}, e_{25} \rangle, \\ H_3 &= \langle a_s, a_t, b_s, b_t \rangle, \\ M_{X,3} &= \langle Z_3, e_{13}, e_{23}, H_3 \rangle, \\ M_{Y,3} &= \langle Z_3, e_{34}, e_{35}, H_3 \rangle, \qquad N_3 = \langle Z_3, H_3 \rangle. \end{aligned}$$

Here $|H_3| = 64$. The orders and complete maximal-index spectra relevant to the obstruction are

A	$\lvert A\rvert$	$\mathcal{I}(A)$
G_3	419904	$\{2, 9\}$
$M_{X,3}, M_{Y,3}$	46656	$\{2, 9, 81\}$
N_3	5184	$\{2, 81\}$

The single, standalone file `G3-verification.g` accompanying this manuscript enters all eight generators in (6) as literal 5-by-5 matrices. Using only standard GAP commands, it checks the group and subgroup orders, the five named maximal inclusions, every maximal-subgroup conjugacy class and its index, coverage by the named subgroups, and an exhaustive top-down search for an increasing chain. It has been run successfully, without modification, in GAP 4.11.1 and GAP 4.16.0. Thus the computation can be checked from the attached file without access to any online repository.

Remark 4. *GAP computations for $p = 3, 5, 7$ reproduce the predicted spectra; for $p = 3$ these are*

$$\{2, 9\}, \qquad \{2, 9, 81\}, \qquad \{2, 81\},$$

and an exhaustive branch search finds no admissible chain. These computations corroborate, but are not needed for, the proof. The standalone $p = 3$ file, complete code, tests, and captured outputs are publicly available in release v1.1.0 of the accompanying repository [3]. *The certificate was reproduced under GAP 4.11.1 and GAP 4.16.0.*

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Code and data availability

The complete computational supplement is publicly available in the verification repository. Release `v1.1.0` is the version corresponding to this manuscript. It includes the standalone verifier and captured output from GAP 4.11.1 and GAP 4.16.0.

Declaration of generative AI and AI-assisted technologies

During this project the author used Anthropic's Claude and OpenAI Codex for mathematical exploration and criticism, literature-search assistance, drafting and debugging GAP code, and manuscript organization and editing. The author checked the mathematical argument, reran the committed tests, and verified the cited references, and takes full responsibility for the content of the article.

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